

Ocean State Forecast (OSF) Datasets (INCOIS)

Executive Summary: INCOIS's *Ocean State Forecast (OSF)* provides daily forecasts of waves, ocean temperature, currents, mixed-layer depth (MLD), and 20°C isotherm depth (D20) for the Indian Ocean [56†L265-L272]. Operational outputs include global/regional wave model results (WAVEWATCH-III) and ocean model results (HYCOM/ROMS). Recent data are openly posted via INCOIS's THREDDS servers in NetCDF format [76†L12-L14] [85†L137-L139]. For example, the RSMC portal lists files like `rsmc_combined_ww3_20260827.nc` (waves) and `RSMC_hycom_20260828.nc` (currents/SST) [76†L12-L14]. The wave forecasts come from a nested WW3 model (1° global, 0.5° Indian Ocean, 0.25° regional, 0.1° coastal grids) [6†L109-L115], while the ocean forecasts use HYCOM v2.3/2.35 (~6.9 km resolution over 20°E–120°E, 43°S–30°N) [6†L123-L129]. These provide 3-hourly forecasts out to ~5–10 days. All recent data (2010–present for waves; ~2015–present for the global ocean model) are public HTTP downloads [85†L137-L139] [85†L141-L143]. Historical archives (pre-2010) are available upon request via the INCOIS Data Requisition Form [84†L69-L78]. Below we list direct download links, file patterns, contents, and usage examples.

Data Sources and Download Links

- **INCOIS RSMC – Global Waves (WW3):** Files named `rsmc_combined_ww3_<YYYYMMDD>.nc` are published daily on INCOIS THREDDS. Each NetCDF contains wave parameters (e.g. significant wave height, mean wave period, wind speed/direction) on nested grids [56†L265-L272] [6†L109-L115]. Example link (1 Aug 2026):
`https://incois.gov.in/thredds/fileServer/osf/ww3/rsmc\_combined\_ww3\_20260827.nc` [76†L12-L14].
Update: Daily (forecasts for next 5–10 days, 3-hourly intervals). **Archive:** 2010–present [85†L137-L139].
Access: Public HTTP (no login).
- **INCOIS RSMC – Global Ocean (HYCOM):** Files named `RSMC_hycom_<YYYYMMDD>.nc` appear daily. Each NetCDF contains surface ocean variables (sea-surface temperature, surface currents, MLD, D20, salinity, etc.) from a global HYCOM run [56†L265-L272] [6†L123-L129]. Example link (2 Aug 2026):
`https://incois.gov.in/thredds/fileServer/osf/currents2/RSMC\_hycom\_20260828.nc` [76†L12-L14].
Resolution: ~1/16° (~6.9 km) on 20°E–120°E × 43°S–30°N [6†L123-L129]. **Update:** Daily (3-hourly forecasts). **Archive:** ~2015–present (operational OSF began ~2010, upgraded later) [85†L141-L143]. **Access:** Public HTTP.
- **INCOIS-BIO-ROMS (Ocean/Carbon Model):** A high-res ROMS output for 1980–2019 (monthly) is available on Zenodo [65†L39-L43]. This NetCDF (`INCOIS_BIO_ROMS.nc`, 1/12°) contains **seasurface** temperature, salinity, dissolved inorganic carbon (DIC), alkalinity, pH, and pCO₂ (monthly climatologies) [65†L39-L43]. **Link:** [Zenodo INCOIS_BIO_ROMS.nc \(4.1 GB\)](#) [65†L39-L43]. **Access:** HTTP (CC-BY 4.0 license) [65†L143-L149].

Other INCOIS products (e.g. coastal WAM/MIKE, assimilated reanalyses, tide data) are documented in INCOIS data holdings and LAS portals, but direct bulk download requires registration or requests.

Variables, Resolution, Frequency

OSF data include all key surface/ocean variables used by fisheries and weather agencies [56†L265-L272] . In particular, INCOIS OSF forecasts provide (i) significant wave heights, periods, wind speed/direction from WAVEWATCH-III [56†L265-L272] , and (ii) sea surface temperature, surface currents (u,v), mixed-layer depth (MLD), and 20°C isotherm depth (D20) from HYCOM [56†L265-L272] . The WW3 model has four nested domains (up to 10 km nearshore) [6†L109-L115] ; the HYCOM covers the tropical Indian Ocean (6.9 km grid) [6†L123-L129] . Each forecast file spans ~5–10 forecast days at 3-hour intervals. File sizes vary: WW3 daily files are typically a few MB (depending on horizons), HYCOM outputs are larger (tens to hundreds of MB due to high resolution and multiple variables).

Spatial/temporal resolution: As noted, WW3 uses global 1°, Indian Ocean 0.5°, regional 0.25°, coastal 0.1° grids [6†L109-L115] . HYCOM uses 1/16° (~6.9 km). Forecast outputs are 3-hourly up to about 240–300 h.

Variables inside files: The exact variable names can vary by version, but typically include at least **SigWaveHeight**, **MeanWavePeriod**, **WindSpeed**, **U_current**, **V_current**, **Sea_Surface_Temperature**, **Mixed_Layer_Depth**, **Depth_20C** (20°C isotherm) [56†L265-L272] [65†L39-L43] . For example, the Zenodo ROMS file lists `TEMP`, `SAL`, `DIC`, `ALK`, `PH`, `PCO2` for surface fields [65†L39-L43] .

Access Methods

- **Public HTTP/THREDDS:** The above RSMC files are fetched via direct HTTP (e.g. using `requests.get()` or `xarray.open_dataset()` on the THREDDS URL). No registration or API key is needed; simply use the known URL pattern (change the date) to download each file.
- **Data Requisition Form:** Older/historical OSF data (pre-2010 or archived runs) require a formal request. INCOIS provides a [Data Requisition Form](#) (see **Data Holdings** page) [84†L69-L78] . Researchers may fill this form to obtain bulk archives.
- **Earth System Data Portal / LAS:** INCOIS operates a Live Access Server (LAS) and ERDDAP for some datasets (register for access) [37†L77-L85] . However, RSMC OSF products are more easily gotten via THREDDS or requisition.

Example Python Code (Download & Read)

For illustration, here is a Python snippet to download one RSMC WW3 file and extract key variables (e.g. wave height, SST, MLD, D20) to CSV:

```
import requests
import xarray as xr

# 1. Download NetCDF file via HTTP
```

```

url = "https://incois.gov.in/thredds/fileServer/osf/ww3/
rsmc_combined_ww3_20260827.nc"
r = requests.get(url)
open("rsmc_combined_ww3.nc", "wb").write(r.content)

# 2. Open with xarray
ds = xr.open_dataset("rsmc_combined_ww3.nc")

# 3. Inspect variables (print names)
print(ds.data_vars)

# 4. Select relevant fields (example variable names)
# (actual names may differ; adjust after inspecting ds.data_vars)
vars = ['Significant_wave_height', 'Mean_wave_period',
        'Sea_surface_temperature', 'Mixed_layer_depth', 'Depth_20']
available = [v for v in vars if v in ds.data_vars]
data = ds[available]

# 5. Optionally select a spatial point or subset (e.g., nearest lat/lon)
# For global data, one might need to loop over lat/lon or use ds.sel()

# 6. Convert to Pandas DataFrame (time x variables)
df = data.to_dataframe().reset_index()

# 7. Save to CSV/Parquet
df.to_csv("osf_extract.csv", index=False)
print("Extracted data shape:", df.shape)

```

Instructions: Adjust the `url` and variable names as needed. You can also use `xarray.open_dataset(url)` directly without saving to disk. For large files or bulk download, consider looping dates and using `xarray.open_mfdataset`.

Alternative Sources & Mirrors

- **Zenodo (INCOIS datasets):** Beyond official servers, some INCOIS model outputs are mirrored on Zenodo. For example, the INCOIS-BIO-ROMS file above is CC-BY 4.0 [65†L143-L149]. The Zenodo repository (by INCOIS authors) also includes other products (e.g. ocean pCO₂) [65†L39-L43].
- **Data.gov.in / MOSDAC / Kaggle:** No major INCOIS OSF archives were found on India's open data portal or other mirror sites. MOSDAC focuses on satellite data, not OSF. Kaggle contains no official INCOIS OSF upload.
- **NOAA/PACIOOS:** Independent global reanalyses exist (e.g. NOAA HYCOM, ECMWF wave forecasts), but INCOIS's RSMC products are specific to the Indian Ocean and not mirrored on international servers.

In summary, the primary download source is the INCOIS THREDDS/HTTP server (see table below). The INCOIS ESSDP and LAS offer metadata and tools but are less straightforward for bulk OSF download.

Licensing & Usage

INCOIS data products are distributed under open licenses. INCOIS publications use **CC-BY 4.0** (e.g. “this article is licensed under a Creative Commons Attribution 4.0 International License” [117†L886-L894]). The Zenodo INCOIS-BIO-ROMS dataset explicitly has a CC-BY 4.0 license [65†L143-L149] . In practice, INCOIS encourages reuse with proper citation. For example, cite INCOIS when using OSF data. No restrictive registration is required for the THREDDS files [76†L12-L14] , but attribution to INCOIS/MoES is requested by their terms.

Dataset Comparison

Source / Product	Direct URL (example)	Format	Key Variables	Resolution	Update	Time Range
INCOIS RSMC – Global Waves (WW3)	https://incois.gov.in/thredds/fileServer/osf/ww3/rsmc_combined_ww3_20260827.nc [76†L12-L14]	netCDF	Significant wave height, wave period, wind, etc. [56†L265-L272]	Global 1°, IO 0.5°, regional 0.25°, coastal 0.1° [6†L109-L115]	Daily forecasts (3h)	2010 pres [85†L139-L143]
INCOIS RSMC – Ocean (HYCOM)	https://incois.gov.in/thredds/fileServer/osf/currents2/RSMC_hycom_20260828.nc [76†L12-L14]	netCDF	SST, surface currents (u,v), MLD, 20°C isotherm [56†L265-L272]	6.9 km (1/16°) over 20°E–120°E, 43°S–30°N [6†L123-L129]	Daily forecasts (3h)	~2010 pres [85†L143-L149]
INCOIS-BIO-ROMS (Zenodo)	Zenodo INCOIS BIO ROMS.nc [65†L39-L43]	netCDF	SST, SSS, DIC, alkalinity, pH, pCO ₂ [65†L39-L43]	1/12° (~9.2 km) over Indian Ocean [65†L39-L43]	Monthly	1980–2019

Download Workflow (Mermaid)

```

graph LR
    A[Identify dataset] --> B["WW3 or HYCOM?"]
  
```

```
B -->|WW3| C[Construct URL: .../osf/ww3/rsmc_combined_ww3_<date>.nc]
B -->|HYCOM| D[Construct URL: .../osf/currents2/RSMC_hycom_<date>.nc]
C --> E[Download NetCDF via HTTP]
D --> E
E --> F[Open with xarray/netCDF4]
F --> G[Extract variables (height, currents, SST, MLD, D20)]
G --> H[Save to CSV/Parquet for analysis]
```

Figure: Workflow for selecting and downloading an INCOIS OSF file, then reading it with Python (e.g. via *requests* and *xarray*) and extracting key parameters.

Sources: Official INCOIS documentation and portals [76†L12-L14] [6†L109-L115] [6†L123-L129] [85†L137-L139] ; INCOIS data holdings [84†L69-L78] [85†L141-L143] ; academic descriptions [56†L265-L272] ; and INCOIS-Zenodo data (license CC-BY 4.0) [65†L39-L43] [65†L143-L149] [117†L886-L894] . These sources provide file URLs, update info, variables, and licensing for the Ocean State Forecast products.
